

SCX Hallucination Inevitability Theorem

SCX

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Abstract

Proves that high-confidence hallucinations are inevitable for any single-model ($M=1$) LLM. Only multi-model auditing ($M \geq 1$) provides exponential detection guarantees.

0.1 LLM Hallucination Inevitability Theorem

119— 120— 121—The same argument gives a token-level impossibility result for hallucination detection. Let X denote the prompt and generation history, let $T \in \mathcal{V}$ be the next token emitted by an LLM, and let 122—

$$123|Q = p_\theta(T | X)124|$$

125—be the model’s own confidence in the emitted token. Let $G(X) \subseteq \mathcal{V}$ denote the latent set of ground-truth-consistent or otherwise admissible next tokens. For factual prompts, $G(X)$ may be a small set determined by the world; for genuinely ambiguous or creative prompts, $G(X)$ may contain many legitimate continuations. 126— 127—

Definition 0.1 (H-Problem: high-confidence hallucination). 128— 129—A token-level hallucination at threshold τ is the event 130—

$$131|\text{Hall}_\tau(X, T)132| = 133|\mathbf{1}\{T \notin G(X), Q > \tau\}.134|$$

135—The single-model H-problem is to infer Hall_τ from the single observation (X, T, Q) generated by one LLM. 136—

137— 138—

Theorem 0.2 (H-Theorem: single-model hallucination lower bound). 139— 140—*Assume the LLM analogue of SCX assumptions A1–A6: prompts are partitioned into latent states $S \in \mathcal{S}$; training-data corruption has state-homogeneous rate at least η on the ambiguous states; and expert/model errors are bounded and state-conditioned. Let $\mathcal{S}_{\text{amb}} \subseteq \mathcal{S}$ be the set of states for which the SCX Theorem 3 construction applies, i.e., a high-confidence false token under a noise world and a high-confidence legitimate token under an ambiguity world induce the same observed law of (X, T, Q) . Define 141—*

$$142|\rho_{\min}143| = 144|\min_{s \in \mathcal{S}_{\text{amb}}} \mathbb{P}(S = s | Q > \tau), 146|$$

147—with $\rho_{\min} > 0$. For a single LLM observation channel ($M = 1$), every detector that preserves legitimate ambiguous continuations has undetected hallucination probability 148—

$$149|\mathbb{P}(\text{Hall}_\tau = 1 | Q > \tau)150| \geq 151|\eta\rho_{\min}.152|$$

153—Equivalently, no single-model inference rule can distinguish high-confidence hallucinations from high-confidence legitimate continuations on the ambiguous state mass. The stronger bound with total ambiguous mass is obtained by replacing $\rho_{\min}$ with $\rho_{\text{amb}} = \mathbb{P}(S \in \mathcal{S}_{\text{amb}} | Q > \tau)$. 154—

155— 156—

Proof sketch. 157—Fix an ambiguous state $s \in_{\text{amb}}$. By SCX Theorem 3, there are two data-generating processes with identical observable distribution: 158—

$$159|\mathcal{P}_{\text{noise}}(X, T, Q \mid S = s)160| = 161|\mathcal{P}_{\text{amb}}(X, T, Q \mid S = s).162|$$

163—In $\mathcal{P}_{\text{noise}}$, the training continuation is corrupted with probability at least η , so with probability at least η the model emits a token outside $G(X)$ while assigning confidence $Q > \tau$. In $\mathcal{P}_{\text{amb}}$, the same emitted token is a legitimate continuation because the prompt is genuinely ambiguous, so $T \in G(X)$. 164— 165—Any single-model detector is a measurable function of (X, T, Q) . Since the two worlds induce the same law on (X, T, Q) , the detector has the same output distribution in both worlds. If it rejects the token in the noise world, it must also reject the observationally identical token in the ambiguity world. Therefore any detector that does not falsely reject legitimate ambiguous continuations cannot flag the corresponding noisy-world token either. The entire high-confidence corrupted mass in state s is thus undetected. 166— 167—Averaging over high-confidence states gives 168—

$$\begin{aligned} 170|\mathbb{P}(\text{Hall}_\tau = 1 \mid Q > \tau)171| &\geq 172|\sum_{s \in_{\text{amb}}} 173|\eta \mathbb{P}(S = s \mid Q > \tau) \\ 169| &174| \geq 175|\eta \rho_{\min}.176| \end{aligned} \quad 177|$$

178—This is exactly the SCX noise-difficulty indistinguishability argument with “label noise” replaced by “false token” and “intrinsic sample difficulty” replaced by “legitimate prompt ambiguity.” $\square$ 179—

180— 181—

Corollary 0.3 (H-Corollary: inevitability for one model). 182— 183—If $\eta > 0$ and $\rho_{\min} > 0$, high-confidence hallucination is unavoidable for any single-model system ($M = 1$) whose only evidence is its own output distribution. Increasing parameter count can reduce the constants only by reducing the effective η or the ambiguous-state mass; it cannot remove the identifiability obstruction. Within the SCX framework, the escape route is to introduce independent evidence channels—for example, $M > 1$ independently trained or independently audited models, retrieval, tools, or human verification—so that hallucination and ambiguity no longer induce the same observable law. 184—

185— 186—

Theorem 0.4 (H-Bound: independent auditing reduces missed hallucinations). 187— 188—Let $A_m(X, T)$ be auditor m ’s binary flag that $T \notin G(X)$, and set 189—

$$190|Z_m = A_m(X, T), 191| \quad 192|\bar{Z}_M = \frac{1}{M} \sum_{m=1}^M Z_m.193|$$

194—Assume conditional independence across auditors given (X, T, S) and a state-conditioned separation margin $\Delta > 0$ such that, for a threshold θ , 195—

$$\begin{aligned} 196| &197|\mathbb{E}[Z_m \mid \text{Hall}_\tau = 1, S = s]198| \geq \theta + \Delta, \\ &199|\mathbb{E}[Z_m \mid \text{Hall}_\tau = 0, S = s]200| \leq \theta - \Delta 201| \end{aligned} \quad 202|$$

203—for all relevant states s . The SCX audit rule flags hallucination when $\bar{Z}_M > \theta$. Let $\text{Miss} = \{\bar{Z}_M \leq \theta\}$ and $H_s = \{\text{Hall}_\tau = 1, S = s\}$. Then 204—

$$205|\mathbb{P}(\text{Miss} \mid H_s)206| \leq 207|\exp(-2M\Delta^2).208|$$

209—

210— 211—

Proof sketch. 212—Condition on a hallucinated token in state s . The auditor flags $Z_1, \dots, Z_M$ are independent Bernoulli variables with mean at least $\theta + \Delta$. Hoeffding's inequality gives 213—

$$214 \left| \begin{array}{c} 215 |\P(\bar{Z}_M \leq \theta) 216| \leq 217 |\P(\bar{Z}_M - \mathbb{E}[\bar{Z}_M] \leq -\Delta) 221| \\ 218| \leq 219 |\exp(-2M\Delta^2). 220| \end{array} \right|$$

222—This is the same exponential concentration mechanism as the SCX Theorem 1 consistency bound, now applied to hallucination auditing rather than label-noise detection. $\square$ 223—

224— 225—